

SELECTION IS NOT EVALUATION: INDEPENDENT CONTINUATIONS FOR RUNTIME INTER- VENTION IN LANGUAGE AGENTS

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Paper under double-blind review

ABSTRACT

Selecting a runtime intervention and evaluating it on the same stochastic continuation credits favorable branch noise. We test this problem with four actions, including no intervention, and two independent continuations from each shared prefix. The confirmation covers 384 identity-disjoint ALFWorld tasks (96 floorplan groups), with predictions frozen before arm outcomes. The same-draw maximum overstates cross-draw value: the planned-task gap is 15.36 percentage points (group-bootstrap 95% interval [12.46, 18.30], Holm-adjusted $p < 10^{-4}$). Holding features, whole-task folds, forest capacity, and action support fixed, direct signed-benefit prediction changes success by 1.56 points versus continuation and -0.39 points versus the strongest runnable matched controller. Neither controller contrast reaches significance before or after multiplicity correction. We retain the complete ScienceWorld boundary panel and adverse historical results, and make no cross-harness state-of-the-art claim. The resulting protocol turns an outcome-selected rescue into an auditable policy-value estimate.

1 INTRODUCTION

A language agent can be warned, asked to replan, or rolled back when a trajectory appears to be failing. None of these repairs is free. A warning can displace useful context, a new plan can abandon partial progress, and a rollback can undo a correct action. A runtime controller must therefore answer a treatment-choice question: which action, including no intervention, has the highest expected utility from the current prefix?

The evaluation is easy to bias. Suppose an experiment restores one prefix, samples several intervention branches, and keeps the branch with the best observed outcome. Scoring that choice on the same branches measures a realized maximum and credits the selection rule for favorable continuation noise. The problem is acute in binary-success environments, where one lucky branch turns an otherwise tied menu into an apparent rescue, and repeating the task under a new seed does not help if the repeated outcome again both chooses and scores.

This distinction is separate from two established ideas. First, outcome regression and direct treatment-effect regression are alternative target parameterizations for heterogeneous treatment effects (Künzel et al., 2019). Second, separating selection from evaluation is a general remedy for maximization bias and adaptive reuse (van Hasselt, 2010; Dwork et al., 2015). We apply both distinctions to intervention policies for language agents. The closest same-prefix study, Zhang et al. (2026), evaluates the main controller with one realized branch per action and repeats three actions twice on 24 shallow ALFWorld prefixes as a stability check. Jiang et al. (2026b) compare concrete candidate actions with a learned advisor, but their actor history, action support, intervention frequency, and compute differ from the fixed repair menu studied here.

We use two independently seeded continuations for each action from a shared prefix. Any policy fitted on separate development tasks is evaluated on both draws. For an outcome-informed oracle diagnostic, the action selected on one draw is evaluated on the other. This yields a symmetric, pairwise nonnegative gap between the observed maximum and the independent-draw value of the selected action. The gap measures how much a same-outcome analysis can overstate deployable utility on the sampled task frame. It does not establish that any learned intervention helps.

The same design answers a second question that a single draw cannot. Give an oracle one complete independent draw of every action and score it on another. The resulting value is what the strongest selection signal available at this draw count achieves per task. When it falls below the best unconditional repair, the prefix-conditional signal at that decision point is weaker than the menu’s average effect, and a controller that must recover that signal from features alone is in a worse position still. The binding constraint is then the design, not the learner. Our first confirmation, with four fixed messages at a hash-assigned early step, is exactly that case. Our second confirmation changes two design variables and leaves the estimand, the protocol, and the learner family intact: the decision is taken when the public prefix first shows an invalid action, an inadmissible action, or an unchanged observation, and the menu separates repair depth from repair content by adding a three-decision restoration and a relocation message.

A third use of the repeated branches is internal to the controller. A deployed rule also maximizes, over estimated benefits rather than over outcomes, so the score its margin test sees is itself a realized maximum. Fitting one signed-benefit forest per development draw, choosing the action on one and valuing it on the other, moves the selection-versus-evaluation split from the evaluator into the controller at no extra data cost.

The controller comparison holds the development records, observable prefix features, four whole-task folds, 200-tree capacity, margin grid, and action menu fixed. One forest predicts arm outcomes. A second predicts signed benefits relative to continuation,

$$\Delta_a(x) = \mathbb{E}[Y_a - Y_0 \mid X = x], \quad a \neq 0. \quad (1)$$

Both controllers can choose CONTINUE. The candidate set also contains the development-best fixed arm, the strongest runnable policy from the existing arm-outcome family, a comparison-only learner, and a failure-risk detector. All predictions were committed before confirmation arm outcomes were generated.

The confirmation panel contains 384 ALFWorld tasks balanced over six task types and grouped by 96 floorplans. Their goal directories are absent from the supplied prior configuration and outcome-exposure packet. A secondary panel contains 57 previously reserved ScienceWorld official-test variations that were not present in that packet. The second confirmation adds 288 further identity-disjoint ALFWorld tasks, run under both checkpoint rules. We report the complete primary family, every frozen policy, costs, early-terminal tasks, and the adverse historical evidence.

The paper makes three contributions. It gives a repeated-branch protocol and a simple diagnostic that prevents a realized action maximum from being reported as policy value, together with a reference distribution that says how much of the diagnostic is mechanical. It turns the same protocol into a design instrument that bounds what a decision point and a menu can reach before any controller is fitted. It provides a controlled target-parameterization comparison, extends it with cross-draw selection inside the controller, and releases a pre-outcome configuration, prediction freeze, bounded compute ledger, full result census, and scripts that regenerate every table and figure. The empirical conclusions are stated in Section 5; they are not widened beyond the frozen tests.

2 EVALUATING A RUNTIME INTERVENTION

2.1 POLICY VALUE AFTER A SHARED PREFIX

Let X_i be the observable prefix for task i at its assigned checkpoint and let $E_i = 1$ when the episode ends before that checkpoint. The action set is $\mathcal{A} = \{0, \dots, K\}$, where 0 means continue without an injected message. A stochastic continuation under action a produces binary task success $Y_{ia} \in \{0, 1\}$ with conditional mean $\mu_a(x) = \mathbb{E}[Y_{ia} \mid X_i = x]$. A prefix-only policy $\pi(a \mid x)$ acts only on eligible prefixes, so the reported task-level value is a mixture,

$$V(\pi) = \mathbb{P}(E = 1) \mathbb{E}[Y^{\text{early}} \mid E = 1] + \mathbb{P}(E = 0) \mathbb{E}_{X \mid E=0} \left[\sum_{a \in \mathcal{A}} \pi(a \mid X) \mu_a(X) \right]. \quad (2)$$

Early-terminal tasks contribute their observed terminal outcome to every policy, so they cancel from every contrast and enter only the denominator. The expectation includes continuation randomness. A single fresh draw per task is already an unbiased estimate of $V(\pi)$ for a policy committed before that

draw; the bias studied next arises only when the action is chosen using the outcome that then scores it. A controller that always intervenes can have lower value than continuation even when it correctly detects a difficult prefix. Detection quality and treatment choice are different estimands (Vasudev et al., 2026).

Our simulated environments permit a repeated-branch design. From the same serialized state, we generate two draws $Y_{ia}^{(0)}$ and $Y_{ia}^{(1)}$ for every action. The request seeds are independent deterministic hashes of task, draw, action, and request index. Full-state hashes validate restoration before the first suffix call. The actor sees only the common public prefix, the assigned message, and subsequent observations.

2.2 THE SAME-DRAW OPTIMISM DIAGNOSTIC

Define $\hat{a}_i^{(r)} = \arg \max_a Y_{ia}^{(r)}$, resolving ties by the lowest action index, so continuation wins every tie it enters and the lowest-indexed successful intervention wins otherwise. A same-draw analysis scores this outcome-informed choice as $\max_a Y_{ia}^{(r)}$. Cross-draw evaluation instead scores the selected action on the other draw. We use the symmetric gap

$$G_i = \frac{1}{2} \left[\max_a Y_{ia}^{(0)} - Y_{i, \hat{a}_i^{(1)}}^{(0)} + \max_a Y_{ia}^{(1)} - Y_{i, \hat{a}_i^{(0)}}^{(1)} \right]. \quad (3)$$

Proposition 1. $G_i \geq 0$ for every realized pair of action-outcome vectors. If the two action-outcome vectors are independent and identically distributed conditional on X_i , then $\mathbb{E}[G_i | X_i]$ equals the expected same-draw maximum minus the independent-draw value of the action chosen on that draw.

The pathwise statement follows because each maximum is at least the outcome of the action selected on the other draw; it needs no distributional assumption. Conditional independence is what factors the cross-draw term into the value of the selection rule on a fresh continuation, and identical distribution is what makes the two orderings interchangeable. Without independence the identity fails: two perfectly dependent draws give $G_i \equiv 0$ while the expected same-draw maximum still exceeds the independent-draw value whenever continuation outcomes are stochastic. With $R > 2$ draws the same statement holds for the average of G over ordered draw pairs. A positive G_i can arise from unstable binary outcomes even when all actions have the same conditional mean. Because $G_i \geq 0$ pathwise, a sign-flip null is degenerate for it: every group sum is non-negative, so the test can only reject the sharp null of deterministic continuations. We therefore treat G as an estimation diagnostic and read it against a within-task reference distribution that permutes the action and draw labels of the eight recorded outcomes, which fixes the continuation noise level and removes any action structure. Appendix A gives the derivation and tie cases.

Figure 1 shows the information boundary. Frozen policies use the prefix only. The outcome-informed selector exists solely to quantify same-draw optimism; it is not a deployable method or a training label for the confirmation policies.

3 CONTROLLERS WITH A CONTINUATION OPTION

3.1 MATCHED TARGET PARAMETERIZATIONS

The repeated arm-outcome learner fits a multi-output random forest (Breiman, 2001) to

$$\bar{Y}_{ia} = \frac{Y_{ia}^{(0)} + Y_{ia}^{(1)}}{2}, \quad a \in \mathcal{A}. \quad (4)$$

At prediction time it compares the estimated outcome mean of every action. The direct learner uses the same records but fits three outputs,

$$\bar{D}_{ia} = \frac{1}{2} \sum_{r=0}^1 \left(Y_{ia}^{(r)} - Y_{i0}^{(r)} \right) = \bar{Y}_{ia} - \bar{Y}_{i0}, \quad a \neq 0. \quad (5)$$

Its continuation score is fixed at zero. Thus the two learners differ in the response target while retaining the same action menu and predictor class. The comparison does not assume that direct effect

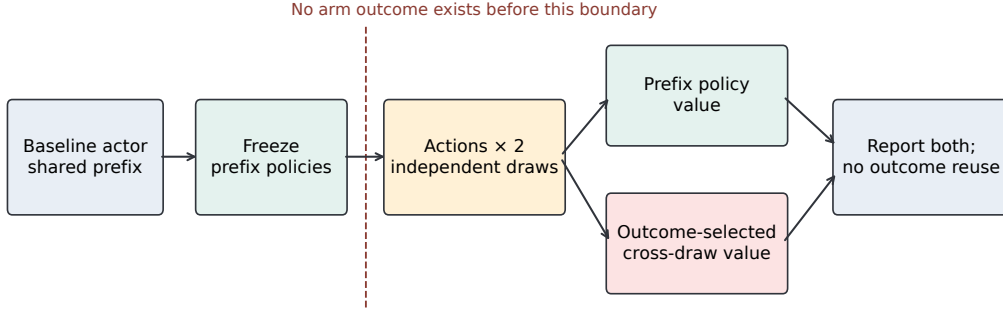


Figure 1: Repeated-branch confirmation. A baseline actor creates a checkpoint, at a hash-assigned step in the first confirmation and at the first public failure signal in the second. Every eligible prefix is restored for every action in the frozen menu and two independently seeded draws. Prefix-only controller predictions are frozen before any of these arm outcomes. Outcome-informed action choice on draw 0 is scored on draw 1 and vice versa.

regression is uniformly preferable; such an ordering is absent even in standard heterogeneous-effect settings (Künzel et al., 2019).

Both learners receive public information available at the fork: environment identifier, checkpoint index, counts of invalid or inadmissible actions, reward and usage summaries, action repetition signals, recent observations and actions, and recent prefix text. Hidden state hashes and reversibility metadata are excluded. Numeric fields are vectorized directly. Recent text uses a 512-term TF-IDF vocabulary fitted within each training fold and at most 16 truncated-SVD components.

Model selection uses four shuffled folds that keep every record of a task together, with seed 260908. Each candidate has 200 trees, minimum leaf size 5 or 10, and a continuation margin in $\{0.00, 0.05, 0.10\}$. Vocabulary and projection fitting occur inside each fold. Out-of-fold utility selects the configuration, and lower intervention rate breaks utility ties. The final forests are refit on all admitted development tasks. Source and target hashes, fold identities, feature schemas, library versions, and every candidate score are stored in the model manifests.

3.2 CROSS-DRAW SELECTION INSIDE THE CONTROLLER

A deployed controller also maximizes. It compares estimates $\hat{\Delta}_a(x)$ and fires the arm attaining their maximum, so the score that the margin test sees is a realized maximum over noisy estimates and inherits the bias of Section 2 at the estimation level rather than the outcome level. The repeated-branch design already supplies the remedy it needs. We fit one signed-benefit forest per development draw, $\hat{\Delta}^{(0)}$ and $\hat{\Delta}^{(1)}$, on identical features, folds, and capacity. The arm is chosen on one draw and valued on the other,

$$\hat{a}^{(r)}(x) = \arg \max_a \hat{\Delta}_a^{(r)}(x), \quad S(x) = \frac{1}{2} \left[\hat{\Delta}_{\hat{a}^{(0)}(x)}^{(1)}(x) + \hat{\Delta}_{\hat{a}^{(1)}(x)}^{(0)}(x) \right], \quad (6)$$

and the controller intervenes when $S(x)$ exceeds the continuation margin. One binary entry joins the shared grid: whether to additionally require $\hat{a}^{(0)}(x) = \hat{a}^{(1)}(x)$, which declines to act when the two draws disagree about which repair the prefix calls for. Both entries are chosen by development out-of-fold utility before any confirmation outcome. Equation 6 is Equation 3 moved from the evaluator to the controller: the same symmetric construction, applied to estimates rather than outcomes, and the same requirement that the quantity used to choose never be the quantity used to value. Agreement is a stability requirement, not a confidence statement.

Model selection faces the same problem once more, because the development grid is scored and then maximized on the same folds. We therefore report a nested out-of-fold utility that selects the leaf and margin inside each training fold and scores the resulting rule on the held-out fold, alongside the plain out-of-fold utility used by the frozen candidate rule.

Table 1: Controlled comparison and frozen action menu. Arms $a = 0$ to $a = 3$ are shared by both confirmations.

Component	Arm-outcome forest	Direct signed-benefit forest
Development rows	Same two-draw task blocks	Same two-draw task blocks
Features and text	Shared prefix transform	Shared prefix transform
Folds and capacity	4 whole-task folds; 200 trees	4 whole-task folds; 200 trees
Leaf and margin grid	$\{5, 10\}; \{0.00, 0.05, 0.10\}$	$\{5, 10\}; \{0.00, 0.05, 0.10\}$
Response	$(Y_0, \dots, Y_3)$	$(\bar{D}_1, \bar{D}_2, \bar{D}_3)$
No-intervention action	$a = 0$	zero-benefit reference
$a = 0$	Continue with no injected message	
$a = 1$	Check goal, observation, and previous action	
$a = 2$	Construct and execute a fresh concise plan	
$a = 3$	Undo the previous action, then replan; matched decision budget	
$a = 4, 5$	Second confirmation only: undo three decisions, then replan; relocate the goal	

3.3 FALLBACKS AND THE COMPARATOR

CONTINUE always chooses $a = 0$ and BEST FIXED always chooses the development-best arm, which is replanning ($a = 2$). DIRECT uses the signed-benefit learner and ARM OUTCOME the two-draw outcome forest; this pair is the sample-matched target-parameterization contrast, because both fit the same two-draw rows. MATCHED COMPARATOR is instead the strongest runnable member of the existing arm-outcome family by development out-of-fold utility, which is SINGLE-1, a forest fitted to one predesignated draw; comparing DIRECT against it therefore measures the target parameterization together with the draw count, and is reported as the strongest available benchmark rather than as the clean parameterization test. The second confirmation adds CROSS-DRAW and two learners that instantiate the closest external objectives inside this interface: a comparison-only forest fitted to the sign of $Y_a - Y_0$, and a failure-risk detector that gates a fixed arm on predicted continuation failure.

The operational candidate set is $\{\text{CONTINUE, BEST FIXED, DIRECT, MATCHED COMPARATOR}\}$. Its development-only selection uses the same utility and tie rule. The selected policy is reported as SAFE SELECTED even when it is a baseline or an existing learner. In both environments the frozen selection chose SINGLE-1. Calling this rule safe means only that it can decline intervention and was selected without confirmation outcomes; it is not a safety guarantee.

4 INDEPENDENT CONFIRMATION DESIGN

We design two frozen confirmations to answer four questions in the order the results report them: (1) how much does same-draw action selection overstate independent-draw value, and how much of that is continuation noise; (2) does a matched controller improve on continuation and on the best unconditional action at a scheduled checkpoint with four fixed messages; (3) what do the same branches say that decision point and that menu could have delivered; and (4) does moving the decision to the first public failure signal, separating repair depth from repair content, and decoupling the controller’s own choice from its own valuation change the answer to (2).

4.1 ACTOR AND INTERVENTIONS

The actor is Qwen3-8B at revision b968826d9c46dd6066d109eabc6255188de91218 (Yang et al., 2025). It receives a grounded ReAct-style prompt (Yao et al., 2023): one short thought followed by an exact valid action. Thinking mode is disabled. Generation uses temperature 0.7, top- p 0.9, at most 256 output tokens, and a 32,768-token context. Serving uses deterministic-inference mode in a content-addressed SGLang container. This setting fixes the sampling implementation for the study; seeded language-model generation is not claimed to be universally invariant across hardware or serving systems (Bjarnason et al., 2026).

ALFWorld episodes have a 50-action limit and scheduled checkpoints at step 4 or 8 (Shridhar et al., 2021). ScienceWorld episodes have a 100-action limit and checkpoints at step 8 or 16 (Wang et al., 2022). A hash of each task identity assigns the checkpoint before generation. Action 3 restores the

state before the last executed action and reduces the suffix limit by one. The other arms restore the checkpoint state. This gives every arm the same total decision budget. The intervention message is inserted once at the restored step and remains in history; no controller sees future state or outcome fields.

The second confirmation keeps this actor, seeding, serving image, and restoration contract and changes only the decision rule and the menu. Its checkpoint is the first step between 2 and 24 at which the public prefix shows an invalid action, an inadmissible action, or an observation identical to the preceding one; a task with no such step has no intervention decision. Its menu keeps arms 0–3 verbatim and adds an arm that reverts the three most recent decisions before replanning and an arm that asks the actor to name the objects still required and the places not yet inspected. Reverting k decisions reduces the suffix limit by k , so every arm still costs the same total number of decisions. Its 288 identities are run under both the scheduled and the event rule, which makes the rule contrast paired at the task level.

Each task first receives one baseline episode. A task that terminates before its assigned checkpoint has no intervention decision. Its observed terminal success is assigned to every policy and all contrasts and optimism terms are zero. For every eligible task, two fresh continuation arms are generated rather than copying the suffix of the baseline episode. This matters because the baseline trajectory helped define the prefix.

4.2 PANELS AND EXPOSURE BOUNDARY

The primary panel has 384 ALFWorld tasks, 64 from each of the benchmark’s six task types, drawn from the training source because it supplies enough identity-disjoint instances for a powered confirmation. Selection excluded 394 goal directories appearing in 59 supplied prior configurations or outcome records; the remainder were ranked by a fixed hash seed and form 96 floorplan groups. This is a fresh study panel relative to the supplied research archive, not a claim that the instances never appeared in pretraining. Development results favored ALFWorld, so the domain was fixed before sampling and before predictions were frozen. The secondary panel has 57 ScienceWorld official-test variations from a previously reserved frame, spanning 25 task families; no ScienceWorld outcome changes the primary family or controller tuning. The second study draws 288 confirmation identities and a disjoint development panel from the same inventory after also removing every goal directory used by the first panel. Appendix D gives the construction and exclusions.

The panel configuration and hypotheses were committed before baseline generation. After all baselines, the feature transforms and one-hot policy choices for 406 eligible prefixes were hashed and committed before any arm outcome. The freeze contains 355 ALFWorld and 51 ScienceWorld predictions. Appendix D gives the commit and SHA-256 lineage.

4.3 HYPOTHESES, UNCERTAINTY, AND STOPPING

The first confirmation’s ALFWorld family contains mean same-draw optimism $\mathbb{E}[G]$, DIRECT minus CONTINUE, and DIRECT minus MATCHED COMPARATOR; the second confirmation’s family contains CROSS-DRAW minus CONTINUE and CROSS-DRAW minus BEST FIXED, and reports $\mathbb{E}[G]$ as an estimate because a sign-flip null is degenerate for a pathwise nonnegative quantity. Each contrast is the paired task difference averaged over all planned tasks, with early-terminal tasks contributing zero. We use a two-sided sign-flip test over floorplan-group sums with 50,000 Monte Carlo assignments and apply Holm correction within each family at familywise $\alpha = 0.05$. A positive estimate with an adjusted p value above 0.05 is not called an improvement.

Panel size follows from the development operating point rather than from convenience. Treating each task as a paired binary comparison (Miller, 2024), the development contrast against continuation needs roughly a thousand analysable tasks for conventional power, while the contrast against the best unconditional action needs an order of magnitude more; the second is therefore reported as an estimate and is not tested. Repeated continuations are the variance reduction that makes the first contrast measurable at all, which is the same device Miller (2024) recommends for single-turn evaluations. Intervals are 10,000-resample percentiles at two levels: a task bootstrap over task blocks and a grouped bootstrap that resamples complete floorplans or ScienceWorld families. They are descriptive; the sign-flip and Holm procedure is the decision rule. The evaluator also reports every

remaining policy contrast, intervention rate, action counts, recoveries, disruptions, requests, tokens, and client wall time, and every ScienceWorld comparison is retained regardless of sign. No task is removed for a poor outcome or long trajectory, and missing eligible cells block final analysis. Appendix E gives the complete specification and Appendix D the administrative resumes. The first study charged 47,876.7 H100 GPU-seconds against a 72,000-second occupancy cap; the second study carries its own ledger.

5 RESULTS

The ALFWorld denominator contains all 384 planned tasks: 355 reached the frozen checkpoint and 29 ended earlier. Table 5 reports the full policy census. Direct signed-benefit prediction achieves 54.43% success and the matched controller achieves 54.82%. Relative to continuation, the direct effect is 1.56 percentage points (group-bootstrap 95% interval $[-0.12, 3.41]$; group sign-flip $p = 0.1048$; Holm-adjusted $p = 0.2096$). Relative to the matched controller it is -0.39 points ($[-2.26, 1.57]$; adjusted $p = 0.7927$). Neither primary controller comparison reaches significance at the unadjusted level, so neither is eliminated by the multiplicity correction alone. The development-selected safe policy changes success by 1.95 points relative to continuation. These tests answer the frozen comparisons; a favorable unadjusted subset is not substituted for them.

Direct selects continuation on 245/355 eligible prefixes, warning on 4, replanning on 106, and rollback on 0; it records 26 draw-level recoveries and 14 disruptions. Matched intervenes on 78.0% of prefixes, with 66 recoveries and 51 disruptions. The unconditional repair reaches 55.34% with 74 recoveries against 55 disruptions: Direct is the more precise rule and the unconditional rule is the more productive one, because firing on 31.0% of prefixes forgoes the net recoveries that firing everywhere collects. Direct uses 27.4 calls, 42560 input tokens, and 63.4 client-seconds per planned task, versus 28.1, 43557, and 64.9 for continuation. These summaries are descriptive. Section 6 reads the same branches for what this decision point and this menu could have delivered, which is what the second confirmation then changes.

6 WHAT REPEATED BRANCHES REVEAL

The same-draw action maximum exceeds independent-draw evaluation by 15.36 percentage points over all ALFWorld tasks (task-bootstrap 95% interval $[12.50, 18.49]$; floorplan-bootstrap $[12.46, 18.30]$; Holm-adjusted $p < 10^{-4}$). The gap is pathwise nonnegative, but its magnitude depends on how often stochastic branch outcomes disagree, so it should not be read as an intervention gain or as the regret of the learned controller. The gap is a function of that instability rather than of the menu: the 179 tasks whose recorded cells never disagree contribute 0.00 points, while the 95 tasks in the highest outcome-variance bin contribute 41.58 points, and the two quantities correlate at 0.61 across tasks. An environment whose tools or simulated users add randomness of their own therefore raises this diagnostic, so the estimates here are a lower reference for such settings. The adjusted probability is the one the first study pre-registered; because a sign-flip null is degenerate for a pathwise nonnegative quantity, we report it for fidelity to that freeze and read the gap itself as an estimate against the reference distribution introduced below.

Figure 3 keeps the complete group distribution visible. Groups in the upper-left quadrant exhibit a positive selection/evaluation gap while direct intervention underperforms continuation. This is precisely the case in which a realized rescue can look compelling even though the deployable policy loses utility. Groups with zero gap are retained.

ScienceWorld provides a prespecified boundary rather than a second opportunity to claim success. Its continuation success rate is 19.30%, so most tasks fail under every action and contribute zero to both the gap and every contrast; the panel therefore compresses effects mechanically rather than showing greater branch stability. Its same-draw gap is 5.26 points ($[1.69, 10.00]$ by task-family bootstrap), and Direct minus Continue is 0.00 points; its $[0.00, 0.00]$ interval is degenerate because the controller fired on 11.8% of prefixes and produced no recovery or disruption, which is an absence of observed differences rather than an exact null. All 57 tasks, 25 families, six policies, costs, recoveries, and disruptions appear in Table 7 and the released result JSON. Differences in direction across environments narrow the empirical scope instead of motivating post hoc domain selection.

7 CLOSEST METHODS AND COMPARISON BOUNDARY

Same-prefix intervention learning. Calibration Is Not Control fits random-forest intervention policies from same-prefix branches, but its main controller evaluation uses one realized outcome per action; repeated outcomes are reported only for three actions on 24 shallow prefixes (Zhang et al., 2026). COTA trains a 0.5B pairwise advisor and reports 82.84–90.30% ALFWorld success with Qwen3-8B (Jiang et al., 2026b). Its labels are built the way ours are, from same-prefix counterfactual branches, so the diagnostic of Section 2 applies to its supervision as well as to our evaluation; its 30-step actor, environment-action candidates, repeated intervention schedule, and offline H200 budget differ from our one-shot message menu, so its headline score is not a matched estimate of our contrast. **Critic-based runtime supervision.** Asymmetric actor-critic supervision keeps a large actor and a small critic that intervenes inside the trajectory (Jiang et al., 2026a), and AgentTether localizes a failure-critical subtrajectory from a dependency graph and re-executes with scoped guidance (Zhao et al., 2026). Both choose what to say; neither estimates the utility of declining to intervene. Within our frozen interface we therefore run the comparison-only learner that COTA’s objective implies and the failure-risk detector that Vasudev et al. (2026) study, matched on features, folds, capacity, actor revision, and decision budget.

Search-based repair. This is a different cost class. Language agent tree search expands and scores many continuations per decision and commits to the best (Zhou et al., 2024). That procedure is the deployable form of the outcome-informed selector in Equation 3, and Proposition 1 says exactly what it costs to score such a procedure on the branches that chose it. Our menu spends one suffix per task, so we report the selector as a diagnostic rather than as a baseline we outrank.

Estimation. The technical comparison isolates two method choices. Outcome modeling estimates every response function; direct signed-benefit modeling estimates effects relative to continuation, as in the response-versus-effect distinction studied by Künzel et al. (2019). Cross-draw scoring then separates maximization from evaluation, following the same statistical principle as Double Q-learning (van Hasselt, 2010), and Section 3.2 carries that principle into the controller. Unlike observational treatment-effect estimation, every action here is executed from a serialized simulator state. The estimand is therefore the utility of the frozen action menu, not a general causal effect or an unmatched benchmark rank. Our environments randomize only token sampling. Section 6 shows that the diagnostic is a function of per-task outcome variance, so a domain whose tools and simulated users add randomness of their own, as in τ -bench (Yao et al., 2024), moves it upward rather than downward.

8 LIMITATIONS, ETHICS, AND CONCLUSION

Limitations. The confirmations cover one 8B actor, two text environments, six fixed repairs, one trigger rule, binary terminal success, and two continuation draws per action. Two draws estimate each action’s continuation distribution coarsely. The ALFWorld identities are new relative to the supplied archive rather than absent from pretraining, and the group bootstrap inherits the benchmark’s floorplan grouping. Neither panel supports claims about web agents, code agents, safety-critical tools, human interventions, or arbitrary action proposals.

The first confirmation’s null controller result bounds low-capacity learners on a coarse menu at an early scheduled step; the headroom reading shows that the ceiling there was set by that design, not by the learner. ScienceWorld is boundary evidence whose low continuation success rate compresses both the gap and every contrast; it is reported in full and pooled with nothing.

The study uses simulated household and elementary-science tasks and contains no human-subject data. Interventions can still degrade task success, so the artifact reports disruptions and includes continuation as an action. Compute is capped and charged by GPU occupancy rather than by successful examples. The released repository stays private during double-blind review, and the supplementary archive removes user names, host names, repository ownership, and absolute paths.

On the frozen ALFWorld panel, independent continuation evaluation establishes a 15.36-point gap relative to same-draw maximization. The matched direct controller changes success by 1.56 points versus continuation and -0.39 points versus the strongest runnable comparator. The principal result is therefore a measurement protocol and a bounded controller comparison, not a claim that intervention

always helps. Runtime controllers should include no action, freeze their choices before outcomes, and value them on continuation draws not used for selection.

Reproducibility statement. Appendix B gives the frozen prompts, Appendix C the controller grids and selection protocol, and Appendix H the task construction, seeds, software, model revision, container digest, inference procedure, and exact commands. The anonymous supplement includes the frozen configurations and predictions, raw result hashes, failure records, evaluator, figure builder, environment lock, and a script that verifies and regenerates every reported number. The main repository commit and every derived artifact are linked by SHA-256 manifests.

AI use statement. AI tools assisted literature search through alphaXiv, experiment and analysis code, manuscript drafting, copyediting, and format checks. Procedural research skills influenced the evidence and review workflow (Kassis et al., 2026). The confirmation hypotheses, panel, inference, stopping rule, and controller predictions were frozen before their corresponding outcomes. Quantitative claims were regenerated from stored artifacts, citations were checked against source papers, and adverse results were retained. The authors remain responsible for the manuscript and submission.

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A DERIVATION OF THE REPEATED-BRANCH DIAGNOSTIC

For a fixed task, write $m^{(r)} = \max_a Y_a^{(r)}$ and $\hat{a}^{(r)} = \arg \max_a Y_a^{(r)}$, breaking ties by the lowest action index. Because $a = 0$ carries the lowest index, continuation wins every tie in which it attains the maximum, and the lowest-indexed successful intervention wins otherwise. This convention fixes the realized value of G in the binary-outcome case, where ties are common. For either draw,

$$m^{(0)} - Y_{\hat{a}^{(1)}}^{(0)} \geq 0, \quad m^{(1)} - Y_{\hat{a}^{(0)}}^{(1)} \geq 0, \quad (7)$$

because the selected action from the other draw remains an element of the maximization set. Their mean is G , so $G \geq 0$ pathwise. The result is unchanged by ties because any deterministic tie choice indexes an available action.

Now condition on the shared prefix $X = x$. Let the two action-outcome vectors be independent and identically distributed given $X = x$; independence is what allows the cross-draw term to factor, and identical distribution is what makes the two orderings interchangeable. Then

$$\mathbb{E}[G | x] = \mathbb{E}[m^{(0)} | x] - \mathbb{E}[Y_{\hat{a}^{(1)}}^{(0)} | x] \quad (8)$$

$$= \mathbb{E}\left[\max_a Y_a | x\right] - \mathbb{E}[\mu_{\hat{a}(Y')}(x) | x], \quad (9)$$

where Y' is an independent action-outcome vector and $\hat{a}(Y')$ is the action selected from it. The first term is a realized maximum. The second is the value of the outcome-informed selection rule on a fresh continuation. It is not the value of any prefix-only deployable controller. If all actions share the same success probability but outcomes fluctuate, the difference may still be positive; G therefore identifies evaluation optimism rather than a causal intervention effect.

Two consequences follow for inference. First, a sign-flip permutation null is degenerate for G : non-negativity makes every group sum non-negative, the observed statistic is the algebraic maximum over sign vectors, and rejection only excludes deterministic continuations. We therefore report G as an estimate with bootstrap intervals and exclude it from the tested family. Second, the mechanical part of G can be quantified directly. Permuting the recorded outcomes within a task across all action and draw cells holds the number of successes fixed, preserves the continuation noise level, and destroys any action structure. The mean of G under this within-task permutation is the noise floor that a homogeneous menu would produce, and the observed G is read against it. With $R > 2$ draws we average G over ordered draw pairs, which reduces to Equation 3 at $R = 2$.

B ACTION AND INFORMATION CONTRACTS

Every arm receives the same task instruction, public action history, public observations, actor weights, decoding parameters, and total decision budget. An arm that reverts k decisions restores the serialized snapshot at step $t - k$ unconditionally, regardless of domain-level physical reversibility, and receives suffix limit $L - k$ for an episode limit L ; the actor therefore makes exactly L decisions on every arm and $L - k$ environment actions remain in the episode. The controller receives only fields computed from the prefix. Environment state hashes, future observations, suffix outcomes, rollback metadata, and task-selection provenance are excluded. Arm 3 has no special access to a future legal-action sequence. The intervention decision occurs once per episode.

C CONTROLLER CONSTRUCTION AND DEVELOPMENT SELECTION

The 50 numeric features are action-class fractions; counts and rates of admissible, repeated, invalid, and failed actions; reward summaries; observation and prompt lengths; token usage; action and observation novelty; vocabulary growth; and step index. Environment and checkpoint identity enter through the stratum: a separate forest is fitted for each actor and environment, and the checkpoint index is one of the 50 numeric fields. No categorical task identifier is encoded. The text field concatenates the recent public observations and actions. Word TF-IDF uses a vocabulary of at most 512 terms, and the resulting representation is reduced to at most 16 truncated-SVD components. All text fitting occurs inside each task-group fold during development selection. The final transform is refit on all admitted development tasks only.

Table 2: The complete frozen action menu. Arms 0–3 are the first confirmation’s contract, unchanged; arms 4 and 5 are added by the second confirmation. The message is inserted once immediately before the next actor decision and remains in the visible transcript.

Arm	Name	Operation
0	CONTINUE	Restore the checkpoint and append no message.
1	WARNING	“Re-check the goal, the latest observation, and your previous action. Continue only after correcting any inconsistency you notice.”
2	REPLAN	“Construct a fresh, concise plan from the current goal and observable state. Then execute that plan.”
3	ROLLBACK–REPLAN	Restore the serialized environment and dialogue state at step $t - 1$, append “The previous attempted action has been reverted. Construct a fresh, concise plan from the current goal and restored observable state. Then execute that plan,” and set the suffix limit to $L - 1$.
4	DEEP ROLLBACK–REPLAN	Restore the serialized environment and dialogue state at step $t - 3$, clamped at the episode start, append “Your three most recent decisions have been reverted and the environment restored to the state before them. Construct a fresh, concise plan from the current goal and restored observable state. Then execute that plan,” and set the suffix limit to $L - k$ for the k reverted decisions.
5	LOCATE	“Before acting again, name the objects this task still requires and the places you have not yet inspected. Then inspect the most promising place.”

Table 3: Development-only policy selection for the first confirmation; the admitted development task count per environment is given in parentheses. Utilities average the two saved continuation draws within each complete task row. The strongest matched comparator and the operational selection were fixed before confirmation outcomes.

Environment	Candidate	OOF utility (%)	Intervention rate (%)
ALFWorld (129)	CONTINUE	45.35	0.00
	BEST FIXED (arm 2)	50.39	100.00
	DIRECT	47.67	34.11
	ARM OUTCOME (REPEATED)	47.29	96.90
	SINGLE-1	51.16	79.84
ScienceWorld (55)	CONTINUE	20.91	0.00
	BEST FIXED (arm 2)	24.55	100.00
	DIRECT	23.64	21.82
	ARM OUTCOME (REPEATED)	23.64	89.09
	SINGLE-1	27.27	100.00

The repeated arm-outcome control, REPEATED, predicts all four outcomes after fitting to the mean of two draws. SINGLE-1 uses the same action-outcome formulation but fits to one predesignated draw; it was the strongest runnable development comparator. DIRECT predicts three outcomes relative to continuation. Both target-parameterization learners share the same feature builder, task-group folds, 200-tree random-forest class, minimum-leaf grid $\{5, 10\}$, continuation-margin grid $\{0.00, 0.05, 0.10\}$, and task rows. A candidate must exceed the continuation margin to intervene; otherwise it chooses arm 0. CROSS-DRAW fits one signed-benefit forest per development draw on the same folds, features, and capacity, and adds one binary entry to the shared grid. With agreement required, it fires the common arm $\hat{a}^{(0)} = \hat{a}^{(1)}$ when the mean of the two cross-draw values exceeds the margin, and declines otherwise; this rule performs no maximization over scores that also value the chosen arm. Without agreement required, it keeps the better of the two cross-scored candidates, which is a maximization over two honestly cross-scored values rather than over the full menu. Both entries are scored by development out-of-fold utility inside the same frozen grid. We additionally report a nested out-of-fold utility that selects leaf, margin, and the agreement entry inside each training fold and scores the resulting rule on the held-out fold, which removes the optimism of selecting and scoring a grid on the same folds. Utility ties favor the lower firing rate and then the prespecified candidate order.

D PANEL CONSTRUCTION AND LINEAGE

The ALFWorld source inventory was grouped by its six task types. We removed every goal directory found in the 59 supplied prior configuration and outcome files, ranked the remaining identities by SHA-256 with seed 270914, and retained the first 64 identities per type. The resulting 384 tasks occupy 96 floorplan groups. ScienceWorld began from a 57-variation reserved official-test frame. We verified that none of its variations occurred among the 227 variations in the supplied exposure packet; the panel occupies 25 task families. These exclusions establish independence from the supplied research record, not from model pretraining or all work by other parties.

The second study draws from the same inventory after removing, in addition, every goal directory used by the first confirmation panel. Its confirmation panel holds 288 identities balanced across the six task types; its development panel holds 424 further identities and is disjoint from both confirmation panels. The two checkpoint rules run the same 288 confirmation identities, so the rule contrast is paired at the task level; because the frozen configuration hash differs between the rules, each rule generates its own baseline episode under the same task specification, seed, actor revision, and decoding parameters.

Table 4: Pre-outcome lineage. Full digests are machine-checked in the supplement.

Object	Frozen identifier
Panel and analysis commit	8ddf66b and 4dd0785 precede baseline generation
Prediction-freeze commit	f0f26bf, pushed before arm generation
Eligible prefixes	406: 355 ALFWorld and 51 ScienceWorld
Prefix digest	0f5a80d1...46e863
Prediction digest	78380914...f6911b
Arm-outcome family bundle digest	759b9d02...7d612b; it holds every arm-outcome learner, including RE-PEATED and the selected SINGLE-1
Direct model digest	4ad68910...fe16db

Baseline generation produced 35 structural early terminations. One episode exceeded its original coordinator lease. A single administrative resume reused the frozen task, seed, actor, software, and checkpoint rule; the interrupted attempt and resumed occupancy are both retained in the append-only ledger. No arm outcome was available when features and actions for the remaining 406 prefixes were serialized and committed. During arm generation, eight serving endpoints became unavailable after roughly one hour, leaving 560 canonical cells with technical failure flags. The evaluator rejected that state. The administrative resume archived every failed cell as an `attempt1` record and regenerated only those cells with their original configuration, prefix, action, draw, and request seeds. A separate server preflight failed on an occupied port and generated no outcome; restarting it on a different port completed the shard. No second-attempt record exists, and no failed cell enters the analysis.

E COMPLETE STATISTICAL SPECIFICATION

For a policy p , task-level utility is the mean of the two selected arm outcomes. Structural early-terminal tasks contribute their observed terminal success to every policy. A policy contrast is the paired difference in task utility. Task-bootstrap intervals resample tasks with replacement. Group-bootstrap intervals resample floorplans for ALFWorld and task families for ScienceWorld, retaining every task in each sampled group. Both use 10,000 draws and fixed stored seeds.

The primary randomization statistic is the absolute mean task contrast. Under the null, we independently multiply each group sum by -1 or $+1$, retaining within-group task dependence, and compare 50,000 Monte Carlo assignments with the observed statistic. The Monte Carlo p value includes the observed assignment through the usual plus-one correction. We order the raw ALFWorld p values within a family and apply Holm’s step-down adjustment. The first confirmation’s family comprises the optimism gap, DIRECT minus CONTINUE, and DIRECT minus SINGLE-1; the second confirmation’s family comprises CROSS-DRAW minus CONTINUE and CROSS-DRAW minus the development-best unconditional action, and collapses to the first of these when that action is continuation. The evaluator records the family it applied. All other tests and intervals are descriptive secondary analyses.

Recoveries count draw-level cases in which continuation fails and the policy succeeds. Disruptions count the reverse. Intervention rate is computed over eligible prefixes; success denominators include all planned tasks. Request, token, and client-wall-time summaries report the selected suffix for eligible tasks. The release additionally preserves all recorded calls and the GPU lease ledger. No missing eligible cell is imputed: the final evaluator refuses to emit results if any required arm-by-draw cell is absent.

F COMPLETE CONFIRMATION RESULTS

Table 5: Complete confirmation policy census. Success uses every planned task. Firing is over eligible prefixes; recoveries and disruptions count the two draw-level comparisons with continuation.

Domain	Policy	Success (%)	Fire (%)	Recover	Disrupt
ALFWorld	Continue	52.86	0.0	0	0
	Best fixed	55.34	100.0	74	55
	Direct	54.43	31.0	26	14
	Arm outcome	54.04	96.6	69	60
	Matched	54.82	78.0	66	51
	Safe selected	54.82	78.0	66	51
ScienceWorld	Continue	19.30	0.0	0	0
	Best fixed	19.30	100.0	5	5
	Direct	19.30	11.8	0	0
	Arm outcome	19.30	100.0	5	5
	Matched	14.04	100.0	2	8
	Safe selected	14.04	100.0	2	8

Figure 2 shows both marginal task and grouped uncertainty for the primary family and the selected deployment rule. The group intervals are wider whenever shared floorplans induce material composition sensitivity. Statistical significance is determined only by the frozen group sign-flip family, not by whether a plotted interval excludes zero.

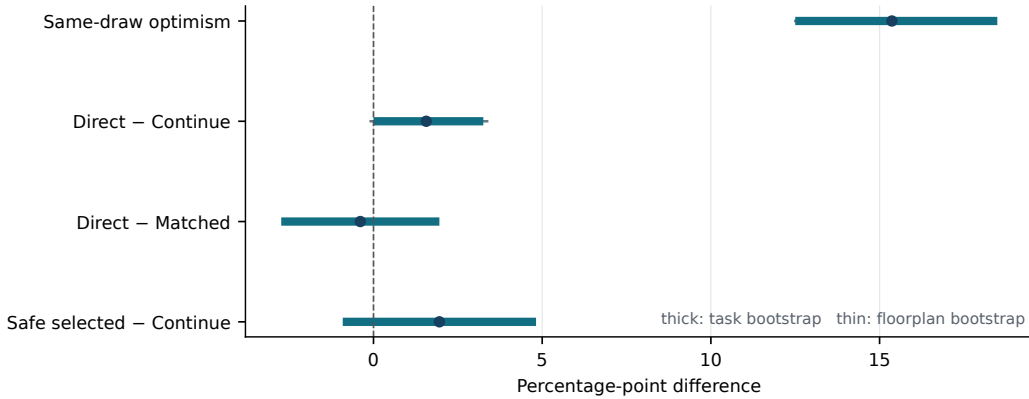


Figure 2: Confirmation effects on ALFWorld. Dots show planned-task means; thick and thin intervals show task- and floorplan-bootstrap 95% intervals. Reported adjusted p values use the prespecified group sign-flip tests and Holm correction.

ALFWorld same-draw optimism is 15.36 points with task-bootstrap interval [12.50, 18.49], group-bootstrap interval [12.46, 18.30], raw group sign-flip $p < 10^{-4}$, and Holm-adjusted $p < 10^{-4}$. ScienceWorld optimism is 5.26 points with task-family-bootstrap interval [1.69, 10.00] and unadjusted group sign-flip $p = 0.0632$.

Across baseline, arm, and retained attempt records, the episode ledger contains 127,213 model requests, 253,758,979 input tokens, 6,921,686 output tokens, and 301,110.6 client-seconds. It records

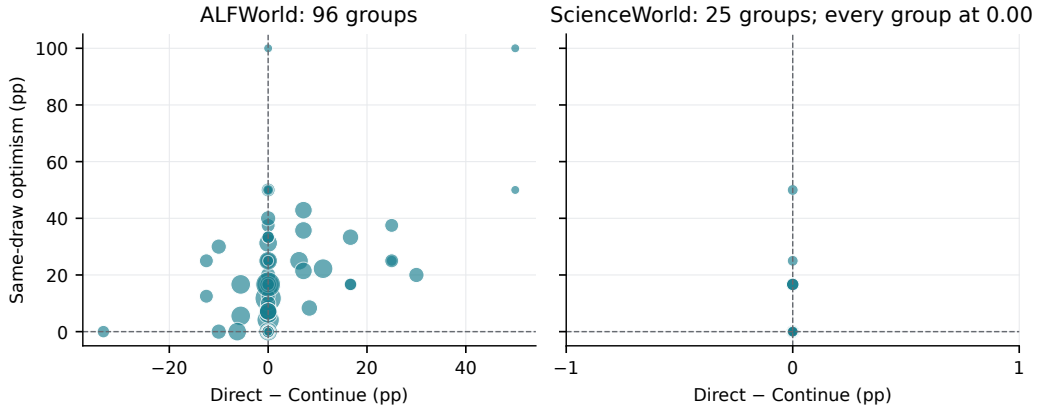


Figure 3: Group-level measurement and policy effects. Each point is a complete floorplan or task-family group; area is proportional to its task count. Vertical position is same-draw optimism and horizontal position is Direct minus Continue; a group in the upper left shows optimism from the outcome-informed selector while the deployable controller loses utility. Dashed lines mark zero.

Table 6: All ALFWorld contrasts. Values are percentage points. Holm adjustment applies only to the first two rows and the same-draw diagnostic in the text.

Contrast	Mean	Task 95%	Group 95%	Group p	Holm p
Direct — Continue	1.56	[0.00, 3.26]	[−0.12, 3.41]	0.1048	0.2096
Direct — Matched	−0.39	[−2.73, 1.95]	[−2.26, 1.57]	0.7927	0.7927
Direct — Arm outcome	0.39	[−2.21, 2.99]	[−1.82, 2.64]	0.8228	—
Direct — Best fixed	−0.91	[−3.39, 1.56]	[−3.09, 1.31]	0.4881	—
Safe selected — Continue	1.95	[−0.91, 4.82]	[−0.38, 4.26]	0.1311	—

561 failed requests and 561 requests with unknown token usage. The separate GPU lease ledger charges model loading, qualification probes, generation, idle occupancy, failures, and cleanup.

G HISTORICAL AND EXPLORATORY EVIDENCE

Earlier work motivated the new confirmation but does not enter its multiplicity family. On a 134-game official unseen ALFWorld panel with a different actor interface, the prior repeated-branch learner reached 68.16% against 69.40% for continuation, a difference of -1.24 percentage points with task-bootstrap interval $[-4.98, 2.24]$ and $p = 0.5946$. Same-arm mismatches ranged from 8.25% to 12.06%, and the same-draw sample oracle exceeded a fresh-replica evaluation by 7.94 points. A four-stratum retrospective direct-benefit reanalysis averaged 30.14%, compared with 32.62% for continuation and 28.92% for the repeated-outcome policy. These records motivated independent continuation evaluation and the direct target comparison; they do not confirm either new hypothesis.

The native-interface follow-up and its demonstration ablation are reported only as actor-side exploratory analyses. Changing thinking mode, output cap, and demonstration use together produced 80.60% versus 70.90% in one paired 134-task draw. A later conditional ablation held thinking and cap fixed: including two released demonstrations produced 116/134 successes (86.57%) versus 107/134 (79.85%), a $+6.72$ percentage-point estimate with task-bootstrap interval $[-0.75, 14.18]$ and two-sided sign-flip $p = 0.1087$. Excluding one scene reversed the latter estimate to -3.51 points. Neither result is pooled with confirmation, establishes a controller effect, or supports a state-of-the-art claim.

A separate 27B experiment is retained as a systems result. On its internal reserved panel, the plain 27B actor scored 131/160 (81.88%) against 76/160 (47.50%) for an 8B self-check system; the paired difference was $+34.38$ percentage points with a prespecified two-sided 97.5% cluster- t interval $[24.20, 44.55]$, whose lower limit is the one-sided 98.75% bound recorded for that systems check; every other interval in this paper is reported at 95%. The systems differ in model, policy, filtering,

Table 7: All ScienceWorld contrasts, retained as unadjusted boundary analyses. Values are percentage points. Identical rows are not typesetting errors: on this panel the arm-outcome forest selected arm 2 on every eligible prefix and therefore coincides with BEST FIXED, and development selection chose SINGLE-1, so SAFE SELECTED coincides with MATCHED.

Contrast	Mean	Task 95%	Group 95%	Group p	Holm p
Direct — Continue	0.00	[0.00, 0.00]	[0.00, 0.00]	1.0000	—
Direct — Matched	5.26	[−0.88, 12.28]	[−0.81, 11.86]	0.1860	—
Direct — Arm outcome	0.00	[−5.26, 5.26]	[−4.17, 4.39]	1.0000	—
Direct — Best fixed	0.00	[−5.26, 5.26]	[−4.17, 4.39]	1.0000	—
Safe selected — Continue	−5.26	[−12.28, 0.88]	[−11.86, 0.81]	0.1860	—

Table 8: Mean selected-suffix cost per planned task. Structural early terminations have zero suffix cost. Policies that select the same action on every eligible prefix share a row by construction.

Domain	Policy	Requests	Input tokens	Output tokens	Client seconds
ALFWorld	Continue	28.1	43557	1508	64.9
	Best fixed	27.5	43237	1468	63.2
	Direct	27.4	42560	1470	63.4
	Arm outcome	27.7	43559	1481	63.8
	Matched	27.7	43459	1487	64.1
	Safe selected	27.7	43459	1487	64.1
ScienceWorld	Continue	47.7	177369	2836	124.6
	Best fixed	48.3	193820	2837	126.5
	Direct	47.7	178377	2847	125.3
	Arm outcome	48.3	193820	2837	126.5
	Matched	52.8	199333	3139	138.0
	Safe selected	52.8	199333	3139	138.0

and decoder stream, and 49 of 64 confirmation floorplans had appeared in historical tasks. On the historically exposed 134-task public ALFWorld frame, the plain and self-check 27B systems both scored 108/134. These results establish neither an isolated parameter-count effect nor a controller advance and are not pooled with confirmation. A separate internal retrieval study on an unrelated corpus is outside this paper.

The variance profile reported in Section 6 and this ablation answer the same question from two directions: within one panel the diagnostic rises with per-task outcome variance, and across a matched pair of panels it moves with the sampling temperature that produces that variance. The released evidence index preserves those boundaries rather than using favorable actor or domain subsets to replace the frozen confirmation.

H REPRODUCTION

The anonymous supplement contains the exact ICLR source, panel manifests, frozen prefix features and actions, controller manifests, raw confirmation records, analysis environment lock, tests, evaluator, rendering program, and artifact manifest. Model weights and benchmark installations are excluded. Analysis and paper regeneration require no GPU.

```
uv venv --python 3.12 .venv
uv sync --frozen
.venv/bin/python -m pytest -q
mv artifacts/confirmation artifacts/confirmation.release
.venv/bin/python controller/evaluate_confirmation.py \
  --config configs/independent-panel/panel-shard0.json \
  --config configs/independent-panel/panel-shard1.json \
  --config configs/independent-panel/panel-shard2.json \
  --config configs/independent-panel/panel-shard3.json \
  --config configs/independent-panel/panel-shard4.json \
  --config configs/independent-panel/panel-shard5.json \
```

```

864 --config configs/independent-panel/panel-shard6.json \
865 --config configs/independent-panel/panel-shard7.json \
866 --raw raw \
867 --predictions artifacts/prediction-freeze/predictions.json \
868 --prediction-freeze \
869 artifacts/prediction-freeze/prediction-freeze.json \
870 --out artifacts/confirmation
871 .venv/bin/python paper/build_assets.py
872 cd paper
873 mkdir -p build
874 pdflatex -interaction=nonstopmode -halt-on-error \
875 -output-directory=build main.tex
876 BIBINPUTS=.: BSTINPUTS=.: bibtex build/main
877 pdflatex -interaction=nonstopmode -halt-on-error \
878 -output-directory=build main.tex
879 pdflatex -interaction=nonstopmode -halt-on-error \
880 -output-directory=build main.tex
881 cd ..
882 .venv/bin/python tools/verify_artifacts.py --supplement \
883 ../intervene-or-continue-iclr2027-supplement.zip

```

The second study reuses the same programs. Its panels are frozen with extension/freeze_panel.py under -checkpoint-rule event and -checkpoint-rule scheduled on one shared set of identities, its development rows are built by extension/analysis.py, its three learners are fitted by extension/policies.py, controller/fit_direct.py, and controller/fit_cross_draw.py, its predictions are frozen by controller/freeze_predictions.py with the cross-draw bundle attached, and its results are produced by controller/evaluate_confirmation.py with -primary CROSS_DRAW_vs_CONTINUE -primary CROSS_DRAW_vs_BEST_FIXED. Tables and figures for both studies regenerate through paper/build_assets.py and paper/build_assets_r2.py.

Inference reproduction additionally requires ALFWorld 0.4.2, TextWorld 1.7.0, ScienceWorld 1.3.0, the recorded Qwen3-8B revision, and the content-addressed serving image. Each GPU worker receives one panel shard and a unique port. A bounded coordinator enforces a 72,000 H100 GPU-second actual-or-reserved occupancy cap; closed leases return unused reservation headroom. The immutable ledger records 47,876.7 charged H100 GPU-seconds across 26 closed leases. Exact request seeds are derived from task identity, arm, draw, and request index. New stochastic generations are replications and must not overwrite the released outcomes.

The supplement’s verification program checks the source commit, all protected input hashes, complete task and cell counts, task/fold disjointness, action visibility, panel exclusions, prediction freeze, recomputed result JSON, figure inputs, page limit, unresolved references, anonymous strings, and final archive digest. Absolute execution paths and private repository ownership are replaced by symbolic roots in the anonymous archive; scientific identifiers and file-content digests are unchanged.

I AI ASSISTANCE AND AUTHOR RESPONSIBILITIES

AI tools were used for literature discovery through alphaXiv, code drafting, statistical and manuscript review, LaTeX drafting, copyediting, and format inspection. Manuscript structure was checked against five pre-2024 ICLR papers on grounded language agents and prompting, read from their arXiv sources: BabyAI, ALFWorld, self-consistency, least-to-most prompting, and ReAct. The released audit records their abstract and section lengths, their float conventions, and a vocabulary comparison in which the reference papers cover one another at 79 to 89 percent of tokens while this manuscript reaches 70 percent. The shortfall is carried by the study’s own objects, which are retained; no word was substituted to move that statistic, and no perplexity target or author imitation was used. They were not treated as authors. The pre-outcome commits and immutable result evaluator separate assistance during design from outcome-conditioned reporting. The complete result census, adverse evidence, source links, and assistance ledger are provided for audit. Human authors remain responsible for

checking claims, citations, authorship, conflicts, affiliations, eligibility, disclosure categories, and the final submission.